

NCERT Class 8 Science - Chapter 3: "Health: The Ultimate Treasure" in English:

1. Group the diseases shown in the images as communicable or non-communicable.

Diseases: Cold and flu, Typhoid, Diabetes, Asthma, Chickenpox.

Answer:

- **Communicable Diseases:** Cold and flu, Typhoid, Chickenpox.
- **Non-communicable Diseases:** Diabetes, Asthma.

2. Diseases can be broadly grouped into communicable and non-communicable diseases. From the options given below, identify the non-communicable diseases.

(i) Typhoid

(ii) Asthma

(iii) Diabetes

(iv) Measles

(a) (i) and (ii)

(b) (ii) and (iii)

(c) (i) and (iv)

(d) cite_start and (iv)

Answer: (b) (ii) and (iii)

(Explanation: Asthma and diabetes are non-communicable diseases as they do not spread from one person to another through pathogens, whereas typhoid and measles are communicable diseases.)

3. There is a flu outbreak in your school.

Several classmates are absent, while some are still coming to school coughing and sneezing.

(i) What immediate actions should the school take to prevent further spread?

- **Answer:**

- Students showing symptoms should be advised to rest at home until fully recovered.
- Ensure proper classroom ventilation and provide adequate handwashing facilities with soap.
- School premises and frequently touched surfaces should be disinfected regularly.

(ii) If your classmate, who shares the bench with you, starts showing symptoms of the flu, how can you respond in a considerate way without being rude or hurtful?

- **Answer:**

- Politely ask them how they are feeling and gently suggest visiting the school nurse, counselor, or teacher for assistance.
- Kindly request them to wear a mask or cover their mouth and nose with a handkerchief when coughing or sneezing.
- Encourage them to take rest at home and assure them that you will share class notes so they don't fall behind in studies.

(iii) How can you protect yourself and others from getting infected in this situation?

- **Answer:**

- Wash hands frequently with soap and water.
- Wear a mask in crowded areas or close spaces.
- Avoid sharing personal items like handkerchiefs, towels, or water bottles.

4.

Your family is planning to travel to another city where malaria is prevalent.

(i) What precautions should you take before, during, and after the trip?

- **Answer:**

- **Before the trip:** Pack long-sleeved clothing, mosquito repellents, and mosquito nets.
- **During the trip:** Use mosquito nets while sleeping, wear protective clothing, and avoid areas with stagnant/still water.
- **After the trip:** Consult a doctor immediately if symptoms like high fever, chills, or sweating develop, and mention the travel history.

(ii) How can you explain the importance of mosquito nets or repellents to your sibling?

- **Answer:**

- Explain that malaria is caused by a protozoan parasite transmitted through mosquito bites.
- Show them that mosquito nets and repellents create a barrier preventing mosquitoes from biting, thereby keeping us safe from illness.

(iii) What could happen if travellers ignore health advisories in such areas?

- **Answer:**

- Travellers are likely to get infected with malaria, suffering from severe symptoms like high fever, profuse sweating, and periodic chills.
- If vector mosquitoes are present in their hometown, they might inadvertently spread the infection to others upon returning.

5.

Your uncle has started smoking just to fit in with his friends, even though it is well known that smoking can seriously harm health and even cause death.

(i) What would you say to him to make him stop, without being rude?

- **Answer:**
 - Express genuine love and concern for his well-being and health.
 - Gently explain that smoking increases the risk of serious health conditions such as lung cancer, asthma, and cardiovascular issues.
 - Encourage him to adopt healthier stress-relief or social activities, such as sports, exercise, or yoga.

(ii) What would you do if your friend offers you a cigarette at a party?

- **Answer:**
 - Firmly but politely say "NO".
 - Explain that smoking is harmful to health and you choose not to compromise your well-being.

(iii) How can schools help prevent students from indulging in such harmful habits?

- **Answer:**
 - Conduct regular awareness workshops, drives, and campaigns highlighting the hazards of substance abuse.
 - Provide active student counseling to foster positive mental health and guide them on handling peer pressure effectively.

6. Saniya claims to her friend Vinita that "Antibiotics can cure any infection, so we don't need to worry about diseases."

What question(s) can Vinita ask her to help Saniya understand that her statement is incorrect?

Answer:

Vinita can ask Saniya the following questions:

1. *"Do antibiotics work against viral infections like common cold, influenza, or COVID-19?"*
2. *"What happens if we take antibiotics without a doctor's prescription or leave a course incomplete—won't bacteria develop antibiotic resistance?"*

3. *"Don't antibiotics also kill beneficial/good bacteria present in our body alongside harmful ones?"*

7.

Dengue Cases Reported in a Hospital Over One Year:

(i) In which three months were the dengue cases highest?

- **Answer: July (65 cases), August (65 cases), and September (65 cases).**

(ii) In which month(s) were the cases lowest?

- **Answer: January (10 cases)** recorded the lowest number of cases.

(iii) What natural or environmental factors during the peak months might contribute to the increase in dengue cases?

- **Answer:**
 - The peak months (July to September) coincide with the monsoon/rainy season.
 - Rainwater accumulation leads to stagnant water collection, providing ideal breeding grounds for mosquitoes.

(iv) Suggest a few preventive steps that the community or government can take before the peak season to reduce the spread of dengue.

- **Answer:**
 - Drain stagnant water and clear potential breeding sites like discarded tires, open coolers, and flower pots.
 - Spray insecticides and introduce biological control agents like *Gambusia* fish in water bodies.
 - Conduct public awareness campaigns on mosquito control and personal protection measures.

8. Imagine you are in charge of a school health campaign.

What key messages would you use to reduce communicable and non-communicable diseases?

Answer:

- **Key Messages for Communicable Diseases:**
 - *"Wash your hands regularly with soap—keep germs away!"*
 - *"Cover your mouth and nose when coughing or sneezing."*
 - *"Prevent water stagnation—stop mosquito breeding!"*
 - *"Get vaccinated on time for lifelong protection."*
- **Key Messages for Non-communicable Diseases:**
 - *"Eat a balanced diet—reduce excess oil and sugar!"*
 - *"Stay physically active—exercise and practice yoga daily!"*
 - *"Limit screen time—ensure proper sleep!"*

- "Say **NO** to tobacco, alcohol, and smoking—say **YES** to health!"

9. It is recommended that we should not take an antibiotic for a viral infection like a cold, a cough, or flu.

Can you provide the possible reason for this recommendation?

Answer:

- Antibiotics are specifically designed to kill **bacteria** by targeting bacterial cell structures.
- Common cold, cough, and flu are caused by **viruses**.
- Antibiotics are ineffective against viral pathogens.
- Indiscriminate or unnecessary use of antibiotics contributes to **antibiotic resistance**, making future bacterial infections harder to treat.

10. Which disease(s) among the following may spread if drinking water gets contaminated by the excreta from an infected person?

Hepatitis A, Tuberculosis, Poliomyelitis, Cholera, Chickenpox.

Answer:

- **Hepatitis A**
- **Cholera**
- **Poliomyelitis**

(Explanation: These diseases are transmitted via contaminated water and food.)

11. When our body encounters a pathogen for the first time, the immune response is generally low but on exposure to the same pathogen again, the immune response by the body is much more compared to the first exposure.

Why is it so?

Answer:

- When the immune system encounters a pathogen for the first time, it takes time to recognize the pathogen and produce specific antibodies to fight it.
- During this primary response, the immune system forms specialized **memory cells**.
- Upon subsequent exposure to the same pathogen, these memory cells instantly recognize the germ and generate a rapid, high-volume secondary antibody response to neutralize it quickly.

- This principle of **acquired immunity** forms the scientific foundation of vaccination.